

CONCEPT CLARITY SHEET

Coding–Decoding & Series and Sequences

Logical & Analytical Reasoning • All Formulas, Rules, Diagrams & Solved Examples

How to use this sheet: Read the concept box, study the diagram, memorise the formula table, then work through the solved examples. Attempt the practice line without looking at the solution first.

A CODING & DECODING

1. What is Coding–Decoding?

Coding is the process of converting a message/word into a secret form (numbers, letters, or symbols) using a specific rule, so it cannot be understood by a third person. Decoding is the reverse — breaking the code to reveal the original message, using the same rule.

Every question gives you one or more coded examples first; your only job is to detect the hidden RULE, then apply it forward (to code) or backward (to decode).

2. The Master Tool — Alphabet Position Chart

This single chart solves almost every coding question. Learn it cold — forward position, backward position, and the opposite-letter pairing.

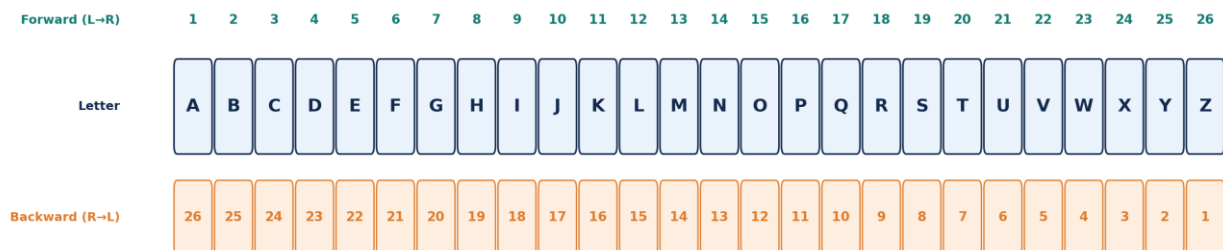


Fig. A1 — Forward position ($A=1 \dots Z=26$) and Backward position ($A=26 \dots Z=1$)

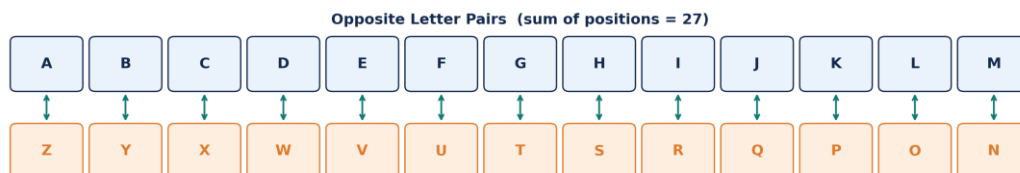


Fig. A2 — Opposite letters always add up to 27 (EBCDIC-style pairing: $A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X \dots$)

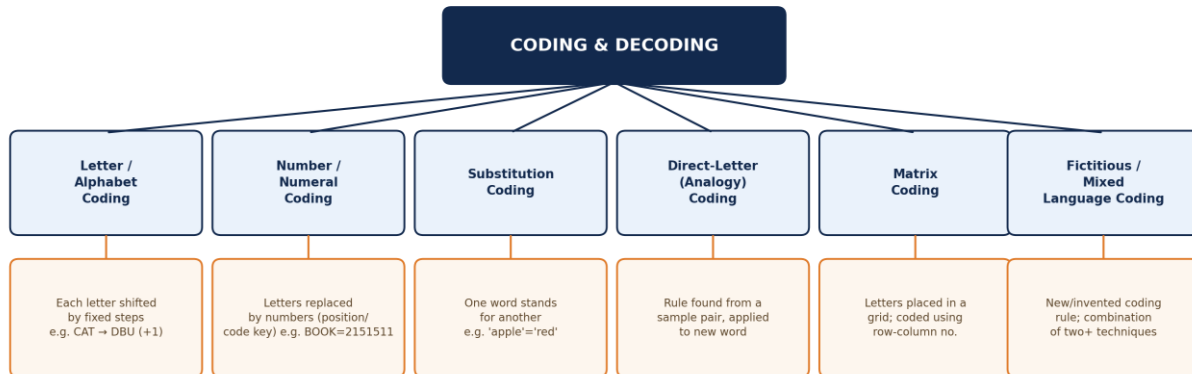
KEY FORMULAS — Coding & Decoding

Forward position	Position of a letter from A $\rightarrow$ value = n (e.g. F = 6th letter)
Backward position	Position from Z $\rightarrow$ value = $27 - n$ (e.g. F $\rightarrow 27 - 6 = 21$)
Opposite letter	Opposite(letter at position n) = letter at position $(27 - n)$
Letter after shift +k	New position = $n + k$ (if > 26 , subtract 26 — wrap around)
Letter before shift -k	New position = $n - k$ (if < 1 , add 26 — wrap around)

Number coding (place value)

Code number = forward position of each letter, concatenated

3. Types of Coding — Visual Map



Golden Rule: Identify the LOGIC on the given word first → then APPLY the same logic to the new word/target.

Fig. A3 — Six common formats coding questions are asked in

4. Solved Examples

Example 1 — Direct Letter Shift Coding

If in a certain code, CAT is written as DBU, how is DOG written in that code?

Solution: Each letter has been shifted forward by +1.

$C(3)+1=D(4)$, $A(1)+1=B(2)$, $T(20)+1=U(21) \rightarrow CAT \rightarrow DBU \checkmark$

Apply same rule to DOG: $D(4)+1=E(5)$, $O(15)+1=P(16)$, $G(7)+1=H(8)$

Answer: DOG $\rightarrow$ EPH

Example 2 — Opposite Letter Coding

In a code language, MOBILE is written as NLYROV. How is TABLET written in the same code?

Solution: Check opposite-letter pairing (sum of positions = 27).

$M(13) \rightarrow 27-13=14=N$, $O(15) \rightarrow 12=L$, $B(2) \rightarrow 25=Y$, $I(9) \rightarrow 18=R$, $L(12) \rightarrow 15=O$, $E(5) \rightarrow 22=V \rightarrow$ each letter mapped to its opposite.

Apply to TABLET: $T \rightarrow G$, $A \rightarrow Z$, $B \rightarrow Y$, $L \rightarrow O$, $E \rightarrow V$, $T \rightarrow G$

Answer: TABLET $\rightarrow$ GZYOVG

Example 3 — Number / Numeral Coding

If BOOK is coded as 2-15-15-11, how will WORD be coded?

Solution: Each letter is replaced by its forward alphabet position.

$B=2$, $O=15$, $O=15$, $K=11 \rightarrow$ matches given code $\checkmark$

$W=23$, $O=15$, $R=18$, $D=4$

Answer: WORD $\rightarrow$ 23-15-18-4

Example 4 — Substitution Coding

In a code language: 'red rose blue' means 'she is happy', 'blue sky red' means 'happy and lucky', 'sky green' means 'lucky day'. What does 'blue' mean?

Solution: Find the common word between sentences that share a common meaning.

Sentence 1 & 2 share word 'blue' and meaning 'happy' → blue = happy.

Answer: blue = happy

QUICK TIPS — Coding & Decoding

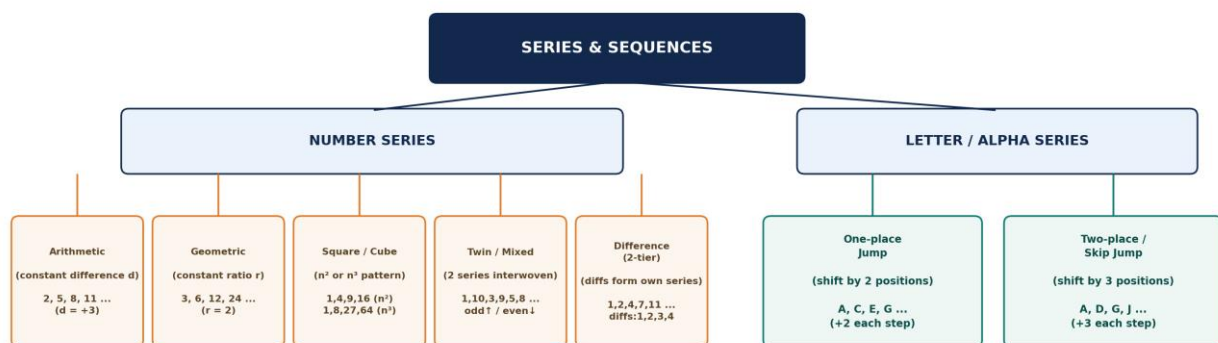
- Always verify your rule on the SAME sample pair given before applying it to the target word.
- Watch for wrap-around: after Z(26), the next letter cycles back to A(1).
- If a letter repeats in the coded word, check letter-count and vowel/consonant patterns too — some questions swap vowels and consonants instead of shifting.
- In number coding, always double-check whether the number represents forward, backward, or reverse-of-word position.
- For substitution/logic puzzles, compare TWO sentences at a time to isolate one word's meaning.

B SERIES & SEQUENCES

1. What is a Series?

A series (or sequence) is an ordered list of numbers or letters that follows one consistent, hidden rule. Series questions ask you to find the next term, a missing term, or the odd term out — by first discovering that rule.

2. Types of Series — Visual Map



Golden Rule: Find the 1st-order difference between terms.
If not constant, find the 2nd-order difference (difference of differences), or check for a $\times/\div$, square, or cube pattern.

Fig. B1 — Number series and letter series, with their common sub-patterns

KEY FORMULAS — Number Series (AP & GP)

Arithmetic Progression (AP) — nth term	$a_n = a + (n - 1)d$ [a = first term, d = common difference]
AP — sum of n terms	$S_n = n/2 \times [2a + (n - 1)d]$

Geometric Progression (GP) — nth term	$a_n = a \times r^{(n-1)}$ [r = common ratio]
GP — sum of n terms	$S_n = a(r^n - 1) / (r - 1)$ [r ≠ 1]
Square series	Term = $n^2 \rightarrow 1, 4, 9, 16, 25, 36 \dots$
Cube series	Term = $n^3 \rightarrow 1, 8, 27, 64, 125 \dots$
nth term via differences	Check 1st differences; if constant $\rightarrow$ AP. If not, take 2nd differences (differences of differences).

KEY FORMULAS — Letter Series	
Letter $\rightarrow$ Number	Use forward position (A=1 ... Z=26) to convert letters to numbers, solve as number series, convert back.
One-place jump	Skip 1 letter each time $\rightarrow$ step of +2 in position value (A, C, E, G ...)
Two-place jump	Skip 2 letters each time $\rightarrow$ step of +3 in position value (A, D, G, J ...)
Alternating series	Two interwoven series at odd & even places — solve each separately.

3. Solved Examples

Example 1 — Arithmetic (AP) Series

Find the next term: 4, 9, 14, 19, 24, ?

Solution: Check the common difference d.

$9-4=5, 14-9=5, 19-14=5, 24-19=5 \rightarrow$ constant difference $d = 5$ (AP)

Using $a_n = a + (n-1)d$, or simply: next term = $24 + 5$

Answer: 29

Example 2 — Geometric (GP) Series

Find the missing term: 5, 15, 45, ?, 405

Solution: Check the common ratio r.

$15/5=3, 45/15=3 \rightarrow$ constant ratio $r = 3$ (GP)

Missing term = $45 \times 3 = 135$ (check: $135 \times 3 = 405 \checkmark$)

Answer: 135

Example 3 — Second-Order Difference Series

Find the next term: 2, 3, 5, 8, 12, ?

Solution: 1st differences are not constant, so find them and then their differences.

Differences: 1, 2, 3, 4 $\rightarrow$ these themselves increase by 1 (2nd-order pattern)

Next difference should be 5 $\rightarrow$ next term = $12 + 5$

Answer: 17

Example 4 — Twin / Alternating Series

Find the next term: 1, 10, 3, 9, 5, 8, 7, 7, ?

Solution: Split into two series at odd and even positions.

Odd positions (1st, 3rd, 5th, 7th...): 1, 3, 5, 7 → increasing by 2

Even positions (2nd, 4th, 6th, 8th...): 10, 9, 8, 7 → decreasing by 1

9th term is odd-position → continues 1, 3, 5, 7 → next = 9

Answer: 9

Example 5 — Letter Series

Find the next term: B, D, G, K, P, ?

Solution: Convert to numbers using forward position and check differences.

B=2, D=4, G=7, K=11, P=16 → differences: 2, 3, 4, 5 (increasing by 1 each time)

Next difference = 6 → $16 + 6 = 22$ → 22nd letter

Answer: V (22nd letter)

QUICK TIPS — Series & Sequences

- First check for a constant difference (AP) — it is the fastest and most common pattern.
- If differences aren't constant, look for a constant ratio (GP) before trying 2nd-order differences.
- Whenever terms grow very fast, suspect squares, cubes, or a $\times 2/\times 3$ pattern rather than addition.
- If a series has 6+ terms and no simple pattern fits, try splitting it into alternate (odd/even position) terms.
- For letter series, always convert to numbers first — it turns an unfamiliar problem into a familiar number series.

60-SECOND RECALL

Coding: Position $n \leftrightarrow (27-n)$ opposite • Shift = $\pm k$ on position • Convert letter to number using A=1...Z=26.

Series: Check constant difference (AP) → constant ratio (GP) → 2nd-order difference → squares/cubes → alternating split.